

Modern Physics

Nature of Light:

→ Light behaves as Ray: $\lambda_{light} \ll \lambda_{object}$

Light behaves as Wave: $\lambda_{light} \approx \lambda_{object}$

Light behaves as particle: when interact with relativistic particles

(e, p, n, α -particle) ←

→ Light particle is called 'photon'

$$E_{photon} = hf = \frac{hc}{\lambda} = \frac{2 \times 10^{-25} J}{\lambda(m)} = \frac{12400 eV}{\lambda(\text{\AA})}$$

($h = 6.626 \times 10^{-34} \text{ m}^2 \text{ kg} \cdot \text{s}^{-1}$)

$$E_{light} = nhf = \frac{nhc}{\lambda} \quad (n \rightarrow \text{No. of photons})$$

→ Since 'n' can only be a whole no.,
E_{light} is always quantized.

$$E_{photon} \propto \text{Frequency } (f)$$

$$E_{photon} \propto \frac{1}{\text{Wavelength } (\lambda)}$$

Characteristics of Photon:

→ Rest mass = 0

→ Speed in vacuum = $c = 3 \times 10^8 \text{ m/s}$

→ Speed in medium = $\frac{c}{\mu}$

→ Charge = None (but diff. than neutron, bcz photon can't be charged)

→ Thus, γ can't deviate in EM field, but can deviate in magnet field gravitational

Imp

→ All radiations like EM waves, X rays, γ -rays, IR waves, etc contains photons, thus not deflected by ext. $\vec{E}$ or $\vec{B}$.

Energy of Relativistic particle →

→ Acc. to 1905 Einstein, Rest mass energy of any particle is

$$E_0 = m_0 c^2 \quad (m_0 \rightarrow \text{Rest mass of particle})$$

→ For any moving particle,

$$E_{total} = \sqrt{(KE)^2 + (E_0)^2}$$

$$E_{total} = \sqrt{(pc)^2 + (m_0 c^2)^2} \quad (p \rightarrow \text{momentum})$$



→ For photon

$$\text{Rest mass} = m_0 = 0 \Rightarrow E_0 = 0$$

$$\Rightarrow E = pc = hf \quad (\text{Hence, this is only Total KE as well as Total energy of } \gamma)$$

$$p_{photon} = \frac{hf}{c} = \frac{h}{\lambda}$$

$$m_{photon} = \frac{hf}{c^2} = \frac{h}{\lambda c}$$

NOTE: Total energy & total momentum are conserved in every e⁻-photon interaction, but no. of photons may or may not be conserved.

$$E.g.: (\text{Pair Production}): \gamma + \gamma \rightarrow e^+ + e^-$$

→ Power of a photon is not defined, only beam of photons (light) has power.

Power & Intensity of Light:

$$P = \frac{E_{light}}{t} = \frac{nhf}{t} = \frac{nhc}{\lambda t}$$

$$\text{No. of photons per unit time} = \frac{n}{t} = \frac{P}{hf} = \frac{P\lambda}{hc}$$

→ Intensity is total energy per unit area, per unit time.

$$I_{light} = \frac{E_{light}}{t} = \frac{nhf}{t} = \frac{nhc}{\lambda t}$$

(*) If the source is same $\lambda \rightarrow$ same then $I \propto n$ (No. of photon)

(*) If source is changed, then I also changes due to change in λ .

$$\rightarrow \text{Point source: } I \propto \frac{1}{r^2}$$

$$\text{Line source: } I \propto \frac{1}{r}$$

Radiation Pressure:

(Complete absorp.)

$$\text{Pressure} = \frac{I}{c}$$

$$\text{Pressure} = \frac{I}{c} (2-S)$$

(S → coeff. of absorp.)

(Complete reflec.)

$$\text{Pressure} = \frac{2I}{c}$$

$$\text{Pressure} = \frac{I}{c} (2+\sigma)$$

(σ → coeff. of ref.)

Violet → Red

Frequency (f) → ↓ as

Wavelength (λ) → ↑ as

Photoelectric Effect:

→ Was discovered by Hertz & Leardard but couldn't explain by wave theory

→ Explained by Einstein in 1920

→ PEE is emission of electron when monochromatic radiation of suitable energy hits a metal surface.

(*) **Work Function (ϕ):** →

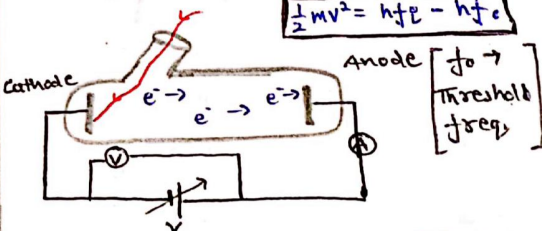
Min. energy required to eject an e⁻ from metal surface.

→ This energy can be given by

- 1) Heat (Thermionic emission)
- 2) Electric Field (Field emission)
- 3) Collision by e⁻ (Secondary emission)
- 4) Light (PEE)

$$KE = E_p - \phi$$

$$\frac{1}{2}mv^2 = hf - hf_0$$



(*) **Necessary conditions for PEE:** →

- 1) Light must be monochromatic
- 2) $E_{light} \geq \phi$ (X) $E_{photon} \geq \phi$ (✓)
(e⁻ may eject, no surffly)

$$3) f_{light} \geq f_0 \quad 4) \lambda_0 \geq \lambda_{light}$$

(*) **Einstein's Equation:** →

→ PEE is one to one interaction

→ It is inst. process (no time lag)

→ Out of several photons, only few are able to find e⁻ & eject them, thus

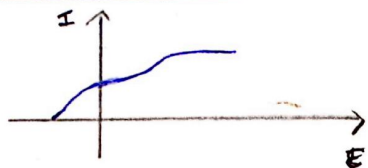
$$\text{efficiency} = \frac{\text{No. of } e^-}{\text{No. of photon}}$$

$$E_{photon} = \frac{hc}{\lambda} = \phi + KE_{max}$$

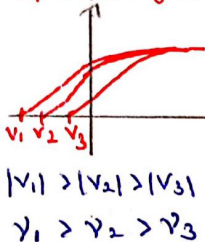
$$\text{Stopping Potential} = eV_0 = KE_{max}$$

Graphs :-

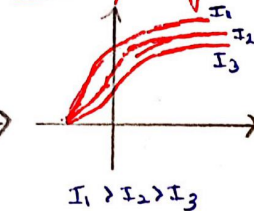
1) Photocurrent vs plate potential $\rightarrow$



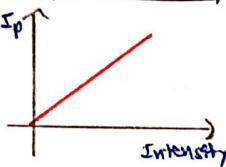
Frequency change



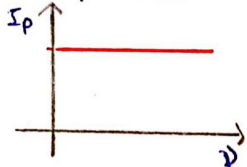
Intensity change



2) I_p vs Intensity



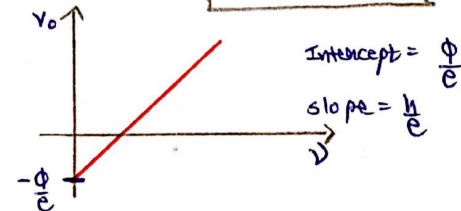
3) I_p vs ν



4) V_0 vs ν

$$E = h\nu = \phi + eV_0$$

$$V_0 = \frac{h\nu}{e} - \frac{\phi}{e}$$



De-Broglie Wavelength :-

$\rightarrow$ He hypothesised that electrons & other matters can behave like matter waves, where λ are as follows

$$\lambda = \frac{h}{p} = \frac{h}{m\psi} \quad (\psi \ll c)$$

$$\lambda = \frac{h}{\gamma m_0 \sqrt{1 - \frac{v^2}{c^2}}} \quad \left[\begin{array}{l} \text{relativistic particle} \\ v \approx c \end{array} \right]$$

$$\lambda = \frac{h}{\sqrt{2m \cdot KE}}$$

$$\lambda = \frac{h}{\sqrt{2m \cdot q \cdot V}} \quad \left[\begin{array}{l} V \rightarrow \text{Potential diff in} \\ \text{elec. field} \end{array} \right]$$

$$\lambda = \frac{h}{m \left(\sqrt{v_0^2 + \frac{qE}{m}} \right)} \quad \left[\begin{array}{l} \vec{E} \rightarrow \\ q \rightarrow v_0 \end{array} \right]$$

$$\lambda = \frac{h}{qBx} \quad (\text{charged particle in B})$$

$$\lambda = \frac{h}{\sqrt{3k_B \cdot T \cdot m}} \quad \left[\begin{array}{l} \text{Gas particle at} \\ T \text{ temp} \end{array} \right]$$

$$\lambda_e = \frac{12.27 \text{ \AA}}{\sqrt{V}} \quad \lambda_p = \frac{0.286 \text{ \AA}}{\sqrt{V}}$$

$$\lambda_0 = \frac{0.202 \text{ \AA}}{\sqrt{V}} \quad \lambda_\alpha = \frac{0.101 \text{ \AA}}{\sqrt{V}}$$

[Deutron is half of α & $h/2$ time of p in mass]

(*) e^- in n^{th} orbit :-

$$\lambda = \frac{h}{\sqrt{2m \left(\frac{13.6}{n^2} \right)}} = \frac{nh}{\sqrt{27.2 \cdot m_e}}$$

$$\Rightarrow \lambda \propto n \text{ (no of orbit)}$$

$\rightarrow$ If a particle at rest decays into two parts of different mass, then also,

$$\frac{\lambda_1}{\lambda_2} = 1 \quad \text{bcz} \quad p_1 = p_2$$

$\rightarrow$ De-Broglie wavelength graph is always rectangular hyperbola, as it is inv. proportional to variables

Photoelectric effect	Compton Effect	Pair Production
Photon p (visible light)	Photon γ (X-ray)	Photon γ (X-ray)
	$\rightarrow$ $E_{\text{photon}} \lambda$ is conserved	$E_{\text{photon}} = 1.02 \text{ MeV}$
	$\lambda_f - \lambda_i = \frac{h(1 - \cos \theta)}{m_e c}$	$e^- \rightarrow \text{electron}$ $e^+ \rightarrow \text{positron}$

X-ray :-

$\rightarrow$ X-ray production is inverse of PE.

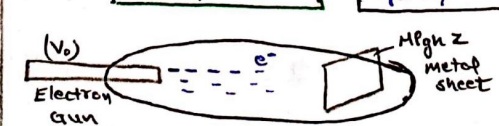
Soft X-ray ($\lambda = 4 \text{ \AA}$) Hard X-ray ($\lambda = 1 \text{ \AA}$)

3) Continuous X-ray :-

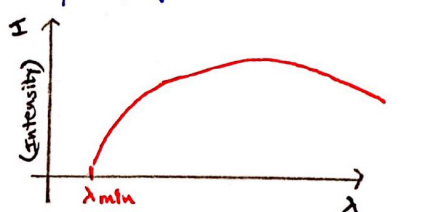
$\rightarrow$ Produced due to retardation of e^-

$$(E)_{\text{max of X-ray}} = (KE)_{\text{max of } e^-}$$

$$\frac{hc}{\lambda_{\text{X-ray}}} = eV_0 \Rightarrow \lambda_{\text{photon}} \propto \frac{1}{V_0}$$

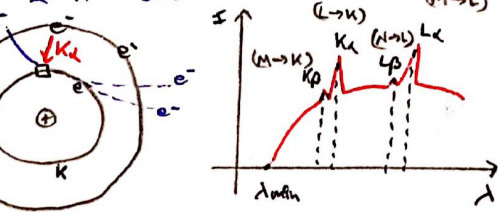


Intensity of X-ray $\propto$ tube current



2) Characteristic X-ray :-

$\rightarrow$ Produced due to collision b/w moving e^- & n^{th} orbit e^- .

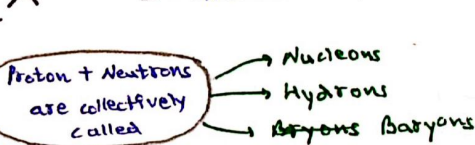


$$\lambda_{\text{min}} \propto \frac{1}{V_0}$$

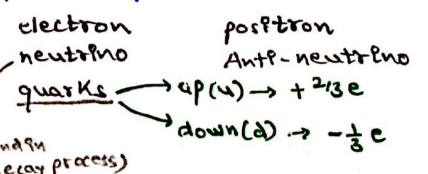
$$\lambda \propto \frac{1}{(Z-1)^2}$$

Nuclei :-

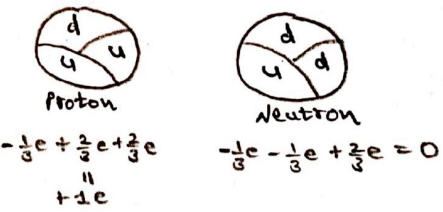
$A = \text{Mass Number}$
 $Z = \text{Atomic Number}$



$\rightarrow$ Here are following fundamental particles of an atom.



$\rightarrow$ Proton & neutron are not fundamental particles. They are made up of quarks.



Mass of subatomic particles :-

$$1 \text{ amu} = \frac{c^2}{12} = 1.66 \times 10^{-27} \text{ kg}$$

$$m_p = 1.67 \times 10^{-27} \text{ kg} = 1.0078 \text{ amu}$$

$$m_n = 1.67 \times 10^{-27} \text{ kg} = 1.0087 \text{ amu}$$

$$m_e = 9.1 \times 10^{-31} \text{ kg} \quad (m_n > m_p)$$

Modern Physics - 2

Rest mass of 1amu :-

$$E = mc^2$$

$$E = 1\text{amu} \times c^2$$

$$E = 931.5 \text{ MeV}$$

Always put mass in amu during energy calculation in nuclear physics

Nuclear size :-

No. of nucleus $\propto A$

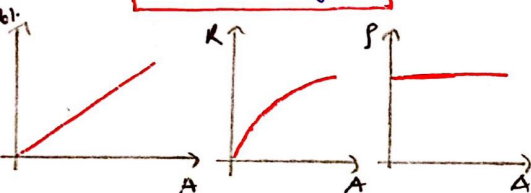
$$R^3 \propto A \Rightarrow R \propto A^{1/3}$$

$$R = R_0 A^{1/3} \quad [R_0 = 1.2/1.4 \text{ fm}]$$

Nuclear Density :-

→ It is same & constant for all the nuclei.

$$\rho = 10^{17} \text{ kg/m}^3$$



Nuclear Force :-

→ Two types of Nuclear forces whose common features are:

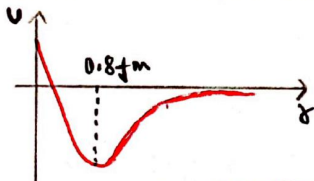
- (i) Non-central
- (ii) Charge independent
- (iii) Short range
- (iv) Does not follow inverse sq. law
- (v) Saturation in nature
- (vii) Ji-ne ki zaroorat, utna hi lagega

Weak Nuclear Force

- Repulsive
- Act in 10^{-16} m
- B/w protons

Strong Nuclear Force

- Attractive
- Act in 1 fm
- B/w both nucleons



No. Nuclear force, when distance b/w nucleons is 0.8 fm

Nuclear Binding Energy :-

→ Energy required to keep nucleus intact is Binding Energy.

$$(\text{Mass of Nucleus}) < (M_p + M_n)$$

This mass diff. is called mass defect.

Jitna zyada mass defect (Δm), utna stable nuclei

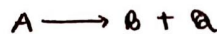
$$\Delta m = Z \cdot m_p + (A-Z)m_n - M$$

(M = Mass of Nucleus)

$$\text{Binding Energy} = \Delta m \times c^2$$

Q value & Binding Energy :-

Consider a Nuclear Rxn



$$Q = [m_A - m_B]c^2 \quad (\text{React} - \text{Product})$$

- $Q = +ve \Rightarrow$ Energy released
- $Q = -ve \Rightarrow$ Energy absorbed

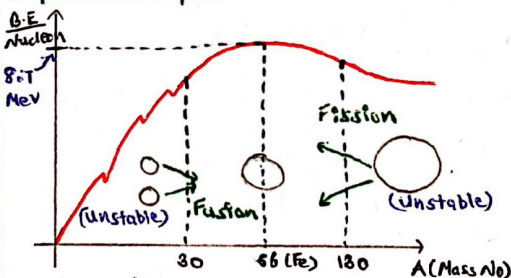
$$Q = (BE)_B - (BE)_A \quad (\text{Product} - \text{React})$$

- $Q = +ve \Rightarrow$ Energy released
- $Q = -ve \Rightarrow$ Energy absorbed

Binding Energy $\Rightarrow$ Not measure of stability

Binding Energy / Nucleon $\Rightarrow$ Measure of stability

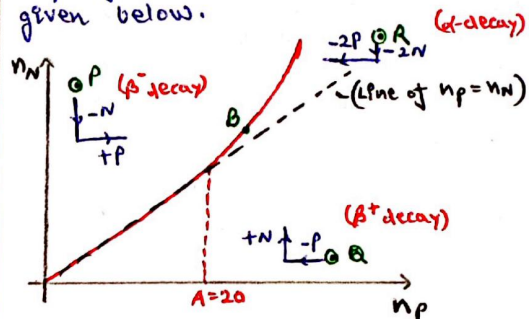
Important Graph :- (1)



- ① $(BE/\text{Nucleon})$ Average is of order of 8 MeV
- ② $\text{Fe}(56)$ is most stable nucleus
- ③ Elements b/w $A=30$ & $A=130$ have stable nuclei
- ④ Elements having $A < 30$ & $A > 130$ are unstable & undergoes fusion & fission.
- ⑤ Energy released in both processes because Unstable $\rightarrow$ Stable.

Imp Graph :- (2)

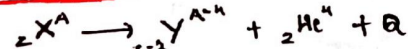
Graph of n_n vs n_p of stable nuclei given below.



- P is unstable due to $(n_n > n_p)$
- A is unstable due to $(n_p > n_n)$
- R is unstable due to $(\text{Mass} \gg 1)$

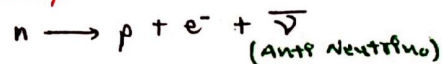
→ upar se neeche β^- decay
neeche se upar β^+ decay

1) α -Decay $\Rightarrow$



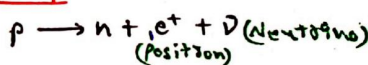
- ① Atomic No $\downarrow$ by 2
- ② Mass No $\downarrow$ by 4
- ③ Helium atom released

2) β^- Decay $\Rightarrow$



- ① Same Mass No.
- ② Atomic No $\uparrow$ by 1

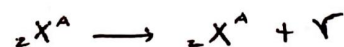
3) β^+ Decay $\Rightarrow$



- ① Same Mass No.
- ② Atomic No $\downarrow$ by 1

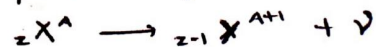
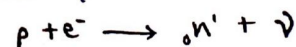
4) γ Decay $\Rightarrow$

- Inside nucleus also, n & p are revolving in orbits.
- After any α or β decay, sometimes vacancy formed in nucleus.
- So outer p goes to inner shell & radiate EMR.



5) e^- capture $\Rightarrow$

- In high atomic no elements, loosely bound p captures nearby e^- to $\uparrow$ stability.



Jitne zyada n , nucleus utna stable, kyonki less p - p repulsion

Law of Radioactivity :

→ Radioactivity is spontaneous decay of unstable nuclei.

→ Radioactivity is random process; which nuclei will decay first can't be predicted.

The activity of a radioactive element at any instant is directly proportional to no. of undecayed nuclei at that time.



$t=0$
 N_0



$t=t_1$
 N

$$\text{Activity} = A = \text{Rate of Decay} = -\frac{dN}{dt}$$

According to law

$$-\frac{dN}{dt} \propto N \Rightarrow -\frac{dN}{dt} = A = \lambda N$$

(λ = Disintegration const.)

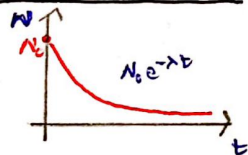
$$N = N_0 e^{-\lambda t} \Rightarrow N = N_0 2^{-t/t_{1/2}}$$

$$A = A_0 e^{-\lambda t} \Rightarrow A = A_0 2^{-t/t_{1/2}}$$

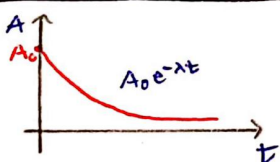
$$(A_0 = \lambda N_0)$$

Imp Points :-

1) N vs time curve →

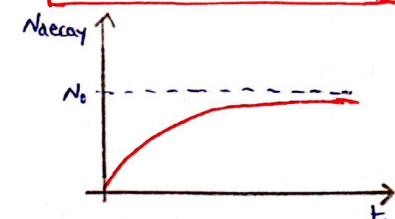


2) A vs time curve →



3) No. of nuclei decayed →

$$N_{\text{decayed}} = N_0 (1 - e^{-\lambda t})$$



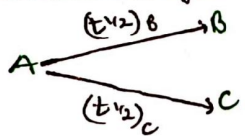
4) Half Life →

$$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

5) Mean Life →

$$t_{\text{mean}} = \frac{1}{\lambda}$$

6) Parallel Disintegration →



$$\lambda_{eq} = \lambda_B + \lambda_C$$

Then

$$\frac{\ln 2}{(t_{1/2})_{eq}} = \frac{\ln 2}{(t_{1/2})_B} + \frac{\ln 2}{(t_{1/2})_C}$$

$$\Rightarrow \frac{1}{(t_{1/2})_{eq}} = \frac{1}{(t_{1/2})_B} + \frac{1}{(t_{1/2})_C}$$

(*) KE of α -particle by Q value :-

① When nuclei at rest undergoes α -decay, stable nuclei & α -particle are released in opp. dirⁿ with same pⁱ.

② Thus, Q value is sum of KE of α -particle & daughter nuclei.

$$(KE)_{\text{daughter}} + (KE)_{\alpha} = Q$$

$$\therefore KE = \frac{p^2}{2m} \Rightarrow \frac{(KE)_{\alpha}}{(KE)_D} = \frac{A-4}{4}$$

$$(\because p_{\alpha} = p_{\text{daughter}})$$

$$KE_{\alpha} = \frac{Q}{A} (A-4)$$

$$KE_D = \frac{Q}{A} (4)$$

$$[A = \text{Mass No. of parent nucleus}]$$

(*) Important Points :-

→ At same energy, penetrating power of particle is

$$\alpha < \beta < \gamma$$

→ Nuclear Fusion → Occurs at very high temp., between smaller nuclei.

Potential energy b/w both nuclei

$$\text{Temp Req.} = \frac{2U}{3K_B}$$

$$\Rightarrow U = \frac{3}{2} K_B T$$

Temp. is so large, that phenomenon

observed only at Sun's surface

③ Energy released per fusion $\approx 20 \text{ MeV}$

→ Nuclear Fission →

↳ can be performed practically
↳ performed for high atomic no nuclei

$$\text{Energy released per fission} \approx 200 \text{ MeV}$$

Power of nuclear reactor →

(i) Nuclear fission is performed in nuclear reactor.

$$(ii) \text{Power (P)} = \frac{NE}{T}$$

N → No. of nuclei undergoing fission in time T

E → Energy released per fission

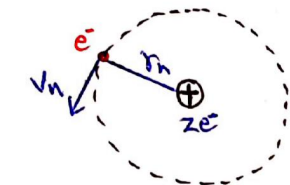
Modern Physics -3 Atomic Physics

Bohr's Postulates :

→ Valid only for hydrogen like single electron species.

1st Postulate ⇒

The centripetal force for revolution of e^- around the nucleus is provided by electrostatic force of attraction.



$$\frac{kZe^2}{r_n^2} = \frac{mv_n^2}{r_n}$$

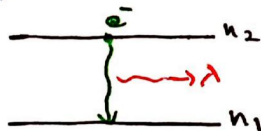
2nd Postulate ⇒

Electrons revolve only in those stable orbits in which their angular momentum is integral multiple of $(h/2\pi)$

$$mv_n r_n = \frac{nh}{2\pi} \quad (n \in \mathbb{I}^+)$$

3rd Postulate ⇒

When electron transition from a higher energy level to a lower energy level, then photon of wavelength (λ) equal to diff. in energy levels is emitted.



$$\frac{hc}{\lambda} = E_{n_2} - E_{n_1}$$

Various Atomic Parameters ⇒

$$\rightarrow r_n = \frac{n^2 h^2}{4\pi^2 kZe^2 m} = 0.529 \times \frac{n^2}{Z} \text{ \AA}$$

$$\hookrightarrow (r_n \propto 1/m)$$

$$\rightarrow v_n = \frac{2\pi kZe^2}{nm} = 2.18 \times 10^6 \left(\frac{Z}{n}\right) \text{ m/s}$$

$$\rightarrow \omega_n = v_n / r_n \quad (\omega_n \propto \frac{Z^2}{n^3}) \text{ rad/s}$$

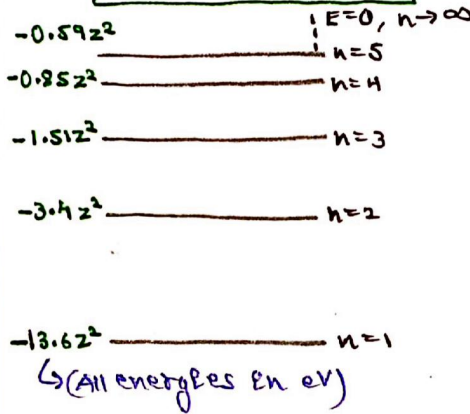
$$\rightarrow T_n = \frac{2\pi}{\omega_n} \quad (T_n \propto \frac{n^3}{Z^2}) \text{ sec}$$

$$\rightarrow E_n = K_n + U_n$$

$$= -\frac{kZe^2}{2r_n} = -13.6 \times \frac{Z^2}{n^2} \text{ eV}$$

Energy level of Hydrogen Atom :

$$E_n = -13.6 \times \frac{Z^2}{n^2} \text{ eV}$$



Excitation of electron :

→ For e^- to absorb energy and excite from n_1 to n_2 , the energy absorbed must be exactly equal to $E_{n_2} - E_{n_1}$

$E_3: -1.51 \text{ eV} \quad n=3$ Electron must absorb 12.09 eV
 $-13.6 \text{ eV} \quad n=1$ to excite from $n=1$ to $n=3$.

λ of emitted radiation :

$$\frac{hc}{\lambda} = E_{n_2} - E_{n_1}$$

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$(R = \text{Rydberg's const.}) = 10^7 \text{ m}^{-1}$$

Also,

$$(a) \lambda = \frac{12430 \text{ \AA}}{\Delta E (\text{eV})}$$

$$(b) \Delta \lambda = 12430 \text{ nm}$$

Number of Spectral lines :

Possible number of photon energies emitted due to de-excitation of electron from $n=n_2$ to $n=1$ state is

$$= {}^n C_2 = \frac{n(n-1)}{2}$$

Hydrogen Spectral Series :

